Before the

FEDERAL COMMUNICATIONS COMMISSION

Washington, D.C. 20554

In the Matter of	)	
	)	
Space Exploration Holdings, LLC	)	ICFS File No.
Application for Authorization of	)	SAT-LOA-20260108-00016
Non-Geostationary Orbit Satellite	)	
System (Orbital Data Center)	)	

PETITION FOR ENVIRONMENTAL REVIEW

PURSUANT TO 47 C.F.R. SECTION 1.1307(c)

AND SUPPLEMENTAL INFORMAL OBJECTION

PURSUANT TO 47 C.F.R. SECTION 25.154

OF NICKOLAI G. BAKKEN

SUMMARY OF THE FILING

This Petition for Environmental Review and Supplemental Informal Objection is submitted pursuant to 47 C.F.R. Section 1.1307(c) and 47 C.F.R. Section 25.154, in support of and supplemental to Petitioner's timely-filed Informal Objection ("Bakken IO") and other pleadings in the SAT-LOA-20260108-00016 docket. It is filed prior to any grant of the Application and requests further environmental processing of SpaceX's Orbital Data Center application for authority to deploy and operate up to one million non-geostationary satellites.

The Petition demonstrates that the present record does not establish the baseline inputs necessary for a reasoned environmental determination, including the annualized steady-state replacement rate, the disposal-method split, the per-unit satellite mass and material profile for each hardware variant, the aggregate reentry mass and material composition of reentering spacecraft, the cumulative atmospheric loading at the described cadence, and the launch cadence required for deployment and steady-state maintenance. At the scale described in the Application, these omissions are not peripheral. They go to whether the Commission can determine, on the present record, that the proposed action will not have a significant environmental effect. The Applicant's own Waiver Requests acknowledge that key operational parameters cannot be accurately described within the current structured format. The Application discloses no recovery, refurbishment, or material-reclamation architecture for end-of-life spacecraft, committing manufactured aerospace hardware to irreversible atmospheric ablation at industrial scale, with a recovery rate of zero. The record does not address whether the non-atmospheric disposal pathways described in the Application are propellant-feasible at the described scale, or whether atmospheric burn-up is the only operationally viable disposal method.

The Petition further identifies record evidence indicating potentially significant environmental effects involving atmospheric contamination from industrial-scale satellite reentry, night-sky and infrared degradation, radio-frequency interference in protected passive bands, orbital-environment transformation, and launch cadence far beyond existing environmental baselines. The cited record includes Murphy et al. (2023), Wing et al. (2026), Maloney et al. (2022, 2025), the Petition to Deny of the American Astronomical Society ("AAS Petition"), the comments of Blue Origin, the Petition to Deny of Amazon LEO ("Amazon LEO Petition"), and Petitioner's own earlier preservation of these issues in the Bakken IO. NEPA identifies irreversible and irretrievable commitments of resources as a required dimension of environmental review and separately requires the study of appropriate alternatives where a proposal involves unresolved conflicts concerning alternative uses of available resources. 42 U.S.C. Sections 4332(2)(C)(iii), (C)(v), and (2)(E). The United States depends on foreign sources for a substantial share of the aerospace-grade alloys and strategic materials used in satellite manufacturing. The present record contains no analysis of cumulative material throughput, no assessment of supply-chain implications, and no evaluation of whether alternative architectures could reduce the irretrievable material loss.

The Petition is filed by an interested person who demonstrates standing under four independently sufficient doctrinal categories, namely: direct user injury arising from reliance on the Applicant's satellite network for connectivity and navigation; aesthetic and environmental injury tied to established night-sky observational practices, atmospheric monitoring, and the documentation of non-repeatable transient astrophysical events; informational injury arising from the deprivation of the factual baseline the Commission's filing framework is designed to produce; and procedural injury arising from documented agency infrastructure failures that prevented timely access to the comment process beginning the day after the Public Notice was released, and from an administrative record that is incomplete, shifting, and no longer current on material dimensions. Environmental review does not occur in isolation from the identity and qualifications of the entity seeking authorization. The environmental certification on Form 312, certifying that the proposed system will have no significant environmental impact, was made under 18 U.S.C. Section 1001 by an entity whose ownership and control picture has materially changed since the certification was executed. On February 2, 2026, the Applicant's corporate parent completed a merger that introduced foreign sovereign-affiliated investors into its shareholder base. Investors in xAI's January 2026 funding round, who became indirect SpaceX shareholders through the all-stock merger, included the Qatar Investment Authority and Abu Dhabi-based MGX, both sovereign-wealth-affiliated entities. See CNBC, "Elon Musk's SpaceX acquiring AI startup xAI ahead of potential IPO" (Feb. 2, 2026) (listing round participants). The Ownership Attachment on file predates this transaction and does not reflect the current ownership picture. The Commission cannot evaluate the reliability of an environmental self-certification without a current and verified understanding of the entity that made it. The Communications Act restricts the grant of any station license to or for the account of a foreign government or its representative, 47 U.S.C. Section 310(a), and the Commission independently reviews foreign ownership in satellite applications under its public interest authority. The Commission's foreign-ownership review obligations and its environmental review obligations are intertwined: both depend on an accurate and current record, and the Application no longer provides one.

The Petition distinguishes the D.C. Circuit's decision in Int'l Dark-Sky Ass'n v. FCC by pointing to the Application's vastly greater scale, materially different altitude and visibility profile, and scientific evidence not present in the Dark-Sky record. The Petition also preserves, without relying exclusively upon, supplemental theories concerning connected actions, anti-segmentation, orbital-commons transformation, Indian religious-site impacts, and aggregate radio-frequency effects.

The Petition requests that the Bureau review the environmental concerns raised, determine that the proposed action may have a significant environmental impact, and require an Environmental Assessment pursuant to 47 C.F.R. Sections 1.1308 and 1.1311, or otherwise employ the appropriate procedural vehicle to ensure these concerns are addressed before any final action on the Application. To the extent this filing is treated concurrently as a petition to deny, the Commission should deny the Application without prejudice to refiling with a complete record, or designate the Application for hearing on the environmental, ownership, and public-interest issues identified herein. In the alternative, the Petition requests a written environmental determination identifying any categorical exclusion relied upon, explaining why the current record does not require further environmental review, and stating on the record the procedural basis for any denial of waiver or alternative treatment, thereby creating a reviewable administrative record. Attachment A supplements the Petition with a Public Interest and Environmental Statement addressing the substantive dimensions the Commission's public-interest analysis must confront on the present record. The evidentiary basis for requiring further environmental review is already on the present record. The six unresolved inputs identified in this Petition are what the Applicant must supply in an Environmental Assessment, not what the Petitioner must supply to trigger one.

Section 1.1307(c) and Section 25.154 are the primary vehicles for this filing. To the extent the Bureau concludes that other Commission rules govern the form, timing, or procedural vehicle for environmental and related objections in this proceeding, Petitioner requests alternative treatment as an environmental informal objection, a petition for other relief, or a petition to deny, together with waiver under 47 C.F.R. Section 1.3 for good cause shown, including waiver to permit concurrent classification under both Section 1.1307(c) and Section 309(d).

Record Documents and Shorthand

For convenience, the following shorthand is used throughout this filing: “NAR” refers to the Application Narrative; “TECH” refers to the Technical Attachment; “WAIV” refers to the Waiver Requests; “SCHS” refers to Schedule S; “OWN” refers to the Ownership Attachment; “AAS Petition” refers to the Petition to Deny of the American Astronomical Society; “Amazon LEO Petition” refers to the Petition to Deny of Amazon LEO; “Blue Origin Comments” refers to the comments of Blue Origin, LLC; “DarkSky Petition” refers to the March 6, 2026 objection or petition of DarkSky International in this docket; “Bakken IO” refers to the Informal Objection of Nickolai G. Bakken filed March 6, 2026; “PN” refers to Public Notice DA 26-113; and “Dark-Sky” refers to *Int’l Dark-Sky Ass’n v. FCC*, 106 F.4th 1206 (D.C. Cir. 2024). "Stewart Petition" refers to the Petition to Deny of William Stewart filed February 9, 2026 (physical) and February 12, 2026 (digital), and his Consolidated Supplemental Petition to Deny filed March 5, 2026.

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I. FILING BASIS AND PROCEDURAL POSTURE

A. Filing Basis

Nickolai G. Bakken, a United States citizen whose official address of record is in Treasure Island, Florida, and who is currently traveling seasonally in Risaralda, Colombia, respectfully submits this Petition for Environmental Review under 47 C.F.R. Section 1.1307(c) and Supplemental Informal Objection under 47 C.F.R. Section 25.154, in support of and supplemental to his timely-filed Bakken IO and other pleadings in this docket. This filing is submitted prior to any grant of the Application and while the Application remains under active Bureau review.

Section 1.1307(c) provides that any interested person may file a written petition alleging that an otherwise categorically excluded action may have a significant environmental effect. This filing is that petition. Section 1.1307(c) contains no express filing deadline and no cross-reference to the comment-period provisions of 47 C.F.R. Section 25.154. By its terms, it requires no showing beyond interested-person status and an allegation of potentially significant environmental effect. Both requirements are met here. The regulation independently directs Bureau review of environmental concerns raised by petition. To the extent the Bureau nonetheless concludes that 47 C.F.R. Section 1.1313(a), 47 U.S.C. Section 309(d), or 47 C.F.R. Section 25.154 governs the form, timing, or procedural vehicle for environmental objections in this proceeding, Petitioner requests in the alternative that this filing be treated as: (a) an environmental informal objection under 47 C.F.R. Section 1.1313(b); (b) a petition for other relief; or (c) a petition to deny under 47 U.S.C. Section 309(d). To the extent the Bureau declines to process this filing under Section 1.1307(c), it should state the regulatory basis for that determination on the record.

B. Relationship to Prior Filings

The Bakken IO was timely filed in ICFS during the comment period established by PN. That filing identified environmental and supervisory sufficiency as a threshold question at Section II.F and preserved the environmental review issue for full development. The present filing supplements and sharpens those preserved issues with fuller record support, additional scientific authorities, and a more focused presentation of the environmental consequences of the Application's described lifecycle-throughput architecture.

C. Verified Facts and Record-Based Arguments

This filing distinguishes between (1) facts within Petitioner's personal knowledge, which are verified in the accompanying Declaration, and (2) legal arguments, technical analysis, scientific authorities, and inferences drawn from the administrative record and publicly available sources. The latter materials are cited and presented for the Commission's consideration and record-preservation purposes, but are not offered as matters of Petitioner's personal knowledge unless expressly so stated.

D. Standing, Party-in-Interest Status, and Non-Competitor Scope

Section 1.1307(c) requires only that the petitioner be an "interested person." However, to the extent the Bureau evaluates this filing under 47 U.S.C. Section 309(d), Petitioner also establishes party-in-interest status. Petitioner is a citizen of the United States. Petitioner holds no competing satellite application, earth-station application, or spectrum license before the Commission, and does not file this Petition for purposes of competitive delay. This Petition is submitted to ensure lawful environmental governance and procedural integrity proportional to the unprecedented magnitude of the Application. Standing is asserted based on particularized injury arising from Petitioner's direct, recurring, personal use of the resources, services, and environments implicated by the requested authorization, as verified in the accompanying Declaration.

E. Concrete and Particularized Injuries

Petitioner demonstrates standing under four independently sufficient doctrinal categories, each supported by verified facts and each satisfying the requirements of Article III and the Commission's interested-person and party-in-interest standards.

1. Direct User Injury. Petitioner owns a Starlink terminal manufactured by the Applicant and relies on the Applicant's network for connectivity during travel, activating service on an intermittent, as-needed basis, including active subscription periods while located in the USA and internationally over the prior twelve months. The proposed system's reliance on optical crosslinks routed through the existing Starlink constellation, at the scale of one million satellites, creates a concrete risk of congestion and degradation to the network capacity on which Petitioner relies. The Application proposes to place up to one million additional spacecraft in the 500 to 2,000 km altitude regime through which the existing Starlink constellation operates. Collision risk scales nonlinearly with orbital density. AAS Petition at 10. An orbital collision cascade in the operational shells would not selectively affect ODC spacecraft. It would degrade or destroy the Starlink network on which Petitioner depends for connectivity, with no replacement timeline bounded in the record. The Application does not address the risk that unbounded object generation from a constellation of this scale could render the orbital environment in which Petitioner's existing service operates permanently degraded.

2. Aesthetic and Environmental Injury. Under Sierra Club v. Morton, 405 U.S. 727 (1972), aesthetic and environmental well-being are cognizable interests when tied to actual use. Under Animal Legal Defense Fund v. Glickman, 154 F.3d 426 (D.C. Cir. 1998) (en banc), a person who repeatedly engages with a feature suffers cognizable injury when that experience is degraded. Under Friends of the Earth, Inc. v. Laidlaw Environmental Services, 528 U.S. 167 (2000), the injury is to the plaintiff's use and enjoyment, not to the environment in the abstract.

Petitioner regularly engages in structured observation of the night sky with his minor daughter, utilizing applications to identify sky events and maintaining a personal archive of night-sky documentation spanning 2020 through 2026. Petitioner has traveled, alone and with his minor child, to observe non-repeatable celestial events including solar and lunar eclipses, aurora borealis displays, and other transient astronomical phenomena. The direct visual experience of a non-repeatable celestial event degraded by orbiting or deorbiting satellite streaks cannot be repeated, and a digital capture of such an event contaminated by those streaks is a permanent loss to the observer's personal record. The event will not recur under the same conditions. Petitioner routinely monitors publicly available space-weather alerts, ionospheric baselines via Schumann resonance outputs, and Fast Radio Burst (FRB) alert streams. FRBs are non-repeatable events at the moment of detection; interference that causes a missed or corrupted transient event is an irretrievable loss. Petitioner uses publicly available FRB detection data and space-weather outputs as inputs for AI-assisted creative and educational projects, including the conversion of astrophysical transient data into structured audio and musical compositions shared with his minor daughter as part of their observational practice. The degradation of the underlying data stream by aggregate radio-frequency contamination from a million-satellite constellation would impair not only the detection of transient events but the downstream creative and educational uses that depend on the integrity of that data. The foreseeable degradation of non-repeatable observation windows by the optical contamination, continuous reentry cadence, and aggregate noise-floor elevation of a million-satellite constellation constitutes a concrete, particularized injury to Petitioner's established observational practices. The injury is to Petitioner's documented, recurring personal use, not to the environment in the abstract.

3. Informational Injury. Under FEC v. Akins, 524 U.S. 11, 21 (1998), the denial or degradation of access to information that a regulatory framework is designed to make available constitutes a cognizable informational injury. Under Public Citizen v. U.S. Department of Justice, 491 U.S. 440, 449 (1989), the deprivation of information required to be disclosed is itself an injury in fact.

The Commission's Part 25 filing framework requires applicants to submit structured technical data sufficient for public evaluation. The Application's Schedule S encodes three satellites for a system described as up to one million. The Applicant's own Waiver Requests acknowledge that the filing vehicle cannot accurately describe the proposed system. The record does not contain ITU advance publication information, coordination data, or any indication of the status of international frequency coordination for the proposed system under Article 9 of the Radio Regulations. To the extent other parties in this proceeding have independently raised concerns regarding the absence or adequacy of international coordination filings, those concerns reinforce the informational deprivation identified here. The Ownership Attachment on file does not on its face satisfy all applicable Form 312 disclosure requirements for the pre-merger ownership structure, and it has not been updated to reflect the post-merger ownership and control picture, which remains unknown on the current record. These omissions deprive Petitioner of the factual baseline the Commission's framework is designed to provide, and on which meaningful public participation depends.

On February 2, 2026, Space Exploration Technologies Corp. completed an all-stock acquisition of xAI, structured as a share exchange. See CNBC, "Musk's xAI, SpaceX combo is the biggest merger of all time, valued at $1.25 trillion" (Feb. 3, 2026) (reporting on transaction documents). The transaction introduced new shareholders into the corporate parent of the Applicant, including investors in xAI's January 2026 funding round such as the Qatar Investment Authority and Abu Dhabi-based MGX. OWN (disclosing pre-merger ownership as of the filing date); see Bakken IO at 18. The Ownership Attachment on file does not reflect the post-merger shareholder base. The Communications Act restricts the grant of any station license to or for the account of a foreign government or its representative. 47 U.S.C. Section 310(a). The Commission independently reviews foreign ownership and control in satellite applications as part of its public interest determination under 47 U.S.C. Section 309(a). Where a post-filing transaction introduces foreign sovereign-affiliated investment into the corporate parent of the Applicant, and the Ownership Attachment on file does not reflect that transaction, the Commission cannot discharge its foreign-ownership review obligations on the current record. This further deprives Petitioner and the public of the informational baseline necessary to evaluate the Application as it now stands, including the ability to assess whether post-filing advocacy reflects the posture of the entity that certified the record.

4. Procedural Injury. Under Lujan v. Defenders of Wildlife, 504 U.S. 555, 572 n.7 (1992), procedural injury is cognizable when the petitioner has concrete interests affected by the procedural violation and the violation increases the risk of harm to those interests. The petitioner need not meet all the normal standards for redressability and immediacy. The right to petition administrative agencies is protected by the First Amendment. California Motor Transport Co. v. Trucking Unlimited, 404 U.S. 508, 510 (1972).

The Public Notice directed commenters to the International Communications Filing System (ICFS). The Commission's public website organizes its filing systems across separate navigation menus. ECFS, the system through which the public ordinarily submits comments on Commission proceedings, appears prominently under the "Proceedings & Actions" menu. ICFS does not appear under that menu. It is located under "Licensing & Databases," a separate navigation path oriented toward licensees and applicants rather than public commenters. A member of the public seeking to comment on a satellite application proceeding would be directed by the website's own information architecture to ECFS, where the Application did not appear in the dropdown menus or prepopulated forms. The ECFS portal offered a "get form" function for comment submission via ecfs@fcc.gov, which returned DNS delivery errors. When Petitioner contacted the FCC's accessibility line and referenced the ICFS docket number, staff directed Petitioner back to ECFS, which could not process the filing. Accessing ICFS required Petitioner to separately register for a CORES account, a credentialing step not required for ECFS participation and not identified in the Public Notice as a prerequisite to commenting. Even after identifying ICFS as the correct filing system, the portal was reportedly inoperable during the period immediately following the Public Notice, further preventing timely electronic participation. These barriers are documented in contemporaneous correspondence with the Commission beginning February 5, 2026, the day after the Public Notice was released, and are preserved for supplemental filing, including screenshots of the Commission's website navigation structure contemporaneous with the comment period. Petitioner is not the only filer who encountered these barriers. The Petition to Deny of William Stewart, filed by paper mail on February 9, 2026 because the electronic filing system was inaccessible, independently documents that the ICFS portal was nonfunctional during the opening of the comment period. See Stewart Petition at 1. To the extent the January

2026 federal government shutdown affected public access to the Commission's electronic filing portals or reduced the availability of staff assistance during the period in which the Public Notice directed public participation, those effects on Petitioner's ability to exercise petitioning rights are preserved.

Petitioner's participation was delayed by the interaction of the Commission's separated filing systems, nonfunctional email infrastructure, staff misdirection, and undisclosed credentialing requirements, not by lack of diligence. The practical consequence of these barriers was not merely delay. It was the suppression of substantive environmental and public-interest arguments from the early comment period, when they could have informed the Commission's initial review posture, shaped the positions of other commenting parties, and entered the public record in a form accessible to other interested persons. The arguments developed in this Petition and in the Bakken IO, including the disposal arithmetic, the atmospheric science chain, the Dark-Sky distinguishing analysis, and the record-completeness deficiencies, were not available to the Commission or the public in a structured, citable form during the period when the Bureau was receiving and evaluating initial comments. A filing that arrives late in the comment period, or after it, enters a record whose contours have already been shaped without it. The Commission's infrastructure failures did not merely inconvenience the Petitioner. They deprived the proceeding of timely public input on the environmental dimensions of the largest satellite application in Commission history, during the window when that input would have had the greatest effect on the Commission's review and on the positions of other parties.

Beyond the specific system failures documented above, the Commission's satellite licensing process presents structural barriers to pro se participation that compounded the delay. The ICFS filing system requires a CORES registration, a separate login credential, familiarity with filing codes and docket conventions not explained in the Public Notice, and the ability to format and upload documents conforming to Commission practice. The Public Notice references "comments and petitions" and cross-references 47 C.F.R. Section 25.154, but does not explain the distinction between informal objections and petitions to deny, does not describe the differing procedural consequences of each vehicle, does not identify the service requirements or formatting conventions applicable to either, does not explain that ICFS requires a separate CORES registration, and does not provide guidance on how to structure a filing to satisfy the Commission's rules for procedural compliance and substantive preservation of issues.

None of these requirements are identified in the Public Notice or seemingly in any publicly accessible guide oriented toward non-attorney commenters. From the date of the Public Notice through the date of Petitioner's first successful filing, approximately thirty days elapsed, a period consumed not solely by drafting but by identifying the correct filing system, obtaining the necessary credentials, understanding the required procedural vehicle, learning the applicable formatting and service rules, and determining how to structure a filing that would be recognized by the Commission as procedurally compliant and substantively preserving. The Commission's rules expressly permit interested persons to participate without counsel. 47 C.F.R. Section 1.1307(c). Where the Commission's own filing infrastructure and procedural conventions effectively require the equivalent of practitioner-level knowledge not only to submit a comment but to ensure that the comment is filed in the correct system, under the correct procedural heading, in the correct format, with proper service, and with arguments preserved in a manner the Commission will recognize, the gap between the rule's invitation and the system's accessibility is itself a barrier to the meaningful public participation NEPA contemplates. A pro se petitioner who files in good faith should not be required to independently reconstruct the procedural knowledge of a communications bar practitioner in order to exercise a right the Commission's own rules confer. This structural barrier is preserved as an independent basis for good cause and as an additional dimension of the procedural injury identified above.

Separately, the Commission's failure to require an Environmental Assessment for a system implicating the atmospheric, optical, and orbital-environment effects documented in this Petition is itself a procedural injury to Petitioner's concrete aesthetic and environmental interests identified above. Where NEPA requires consideration of environmental effects and the agency proceeds without that consideration, persons whose concrete environmental interests are at stake suffer a cognizable procedural injury. See also American Radio Relay League v. FCC, 524 F.3d 227 (D.C. Cir. 2008) (the Commission must make the basis for its decisions available for meaningful public comment).

F. Causation, Redressability, and Good Faith

Each injury identified above is directly traceable to the Commission's potential authorization of the Application, which is a necessary legal precondition to deployment. Each injury is redressable by the relief sought: requiring an Environmental Assessment under NEPA prevents uninformed authorization, and requiring a complete record cures the procedural deprivation.

Petitioner proceeds pro se and in good faith. Pro se filings are held to less stringent standards than formal pleadings drafted by attorneys. Haines v. Kerner, 404 U.S. 519, 520 (1972). That standard arises in the judicial context, but comparable solicitude is appropriate where a Commission rule expressly permits filing by an interested person without requiring legal representation. The Commission is respectfully asked to evaluate the substance of the concerns raised herein rather than the formal credentials of the person raising them.

G. Waiver and Alternative Treatment

Relationship to the Section 25.154 Comment Period. This Petition is filed under Section 1.1307(c), which contains no cross-reference to the comment-period deadlines of Section 25.154 and operates independently as a vehicle for raising environmental concerns before grant. No waiver of the Section 25.154 filing period is required. To the extent the Bureau determines otherwise, Petitioner requests waiver under 47 C.F.R. Section 1.3 for good cause shown. Under WAIT Radio v. FCC, 418 F.2d 1153 (D.C. Cir. 1969), the Commission may not deny a waiver request without a reasoned explanation addressing the special circumstances presented. Special circumstances here include the February 2, 2026 merger that rendered the Applicant's Ownership Attachment obsolete during the comment period, the emergence of peer-reviewed atmospheric science not present in any prior satellite-licensing record, the absence of any prejudice from consideration of environmental concerns that the Bureau is independently obligated to review under Section 1.1307(c), and documented agency infrastructure failures that delayed the Petitioner's participation from the outset of the proceeding. Beginning February 5, 2026, the day after the Public Notice was released, the Application did not appear in the ECFS dropdown menus through which the public ordinarily submits comments, the ecfs@fcc.gov email address returned DNS delivery failures, and the Commission's own accessibility staff, when contacted for assistance on this docket, directed the Petitioner back to the ECFS system that could not process the filing. Petitioner was required to create a separate ICFS login in CORES to participate. To the extent any aspect of this Petition's timing or procedural posture is attributable to delays arising from the agency's own infrastructure failures, those delays constitute independent good cause for waiver.

Good cause further exists based on: novelty of environmental posture; megaconstellation-scale application; newly developed docket record; preserved timely environmental objection; pre-grant submission; active Bureau review; no prejudice to Applicant; record-completeness concerns; compressed pleading window; and the Applicant's own waiver-dependent posture. This filing is submitted prior to grant, while the Application remains under active Bureau review, and as a supplementation of environmental issues timely preserved in the Bakken IO. No party is prejudiced by consideration of these issues now. Exclusion of this filing on procedural grounds would risk omission of substantial environmental concerns from the administrative record notwithstanding Section 1.1307(c)'s directive that the Bureau review such concerns when raised by petition.

Dual Classification. To the extent the Commission's rules do not expressly contemplate a single filing serving concurrently as a petition for environmental review under Section 1.1307(c) and a petition to deny under 47 U.S.C. Section 309(d), Petitioner requests waiver under 47 C.F.R. Section 1.3 to permit dual classification. This filing raises both environmental concerns cognizable under Section 1.1307(c) and public-interest objections cognizable under Section 309(a). Both arise from the same record. Requiring duplicative filings to raise the same evidence under separate procedural headings would serve no purpose other than form, and the Commission's rules should not be construed to require it. The environmental and public-interest dimensions of this Application are intertwined, and the Commission's disposition should address both on the record. To the extent any of the considerations raised in Attachment A Section IX are deemed outside the scope of Section 1.1307(c), they are independently cognizable under the concurrent petition-to-deny classification requested herein, and under the Commission's independent public-interest obligations before grant under 47 U.S.C. Section 309(a). The

Commission may not grant an application without first determining that the public interest would be served, and that obligation is not confined to matters raised within the comment period.

Other Procedural Rules, Formatting, and Convention. To the extent this filing is deemed not to conform to any other requirement of the Commission's rules or practice regarding procedural vehicle, form, format, length, citation style, supporting declaration, service, or other pleading convention, Petitioner respectfully requests waiver under 47 C.F.R. Section 1.3. Given the scale of the Application, the novelty of the environmental questions presented, the developing record in this proceeding, and Petitioner's pro se status, any such non-substantive defect should be waived or, if curable, addressed through leave to supplement or correct rather than through disregard of the concerns raised herein.

Alternative Procedural Treatment. Section 1.1307(c) is the primary and independently sufficient vehicle for this filing. To the extent the Commission declines to process this filing under that section, Petitioner requests in the alternative that it be treated as an environmental informal objection under Section 1.1313(b), a petition for other relief, or a petition to deny under 47 U.S.C. Section 309(d). To the extent the Commission declines any of these classifications, it should state on the record the regulatory basis for that determination, including whether the ruling rests on timeliness, procedural vehicle, standing, lack of good cause, or some other identified ground, so that the basis for the Commission's procedural disposition is preserved in the administrative record.

II. KEY UNRESOLVED INPUTS: WHAT THE RECORD DOES NOT ESTABLISH

The Commission's environmental framework requires a factual basis sufficient to support a reasoned determination that a proposed action will not have a significant environmental effect. That determination is only as sound as the inputs on which it rests. The present record does not establish the following baseline inputs, each of which bears directly on whether the lifecycle-throughput architecture described in the Application may produce significant environmental consequences:

(i) The annualized steady-state satellite replacement rate for a constellation of up to one million spacecraft, each with a described operational life of approximately five years.

(ii) The disposal-method split between atmospheric reentry, Earth disposal orbits, and heliocentric disposal orbits. The Technical Attachment states that "[d]isposal orbit altitudes have not been selected." TECH at A-12. No propellant budget, mass penalty, or operational-feasibility analysis bounds the non-atmospheric alternatives at the described scale.

(iii) The annual aggregate reentry mass and material composition of reentering spacecraft, including the mass fraction of aerospace-grade materials and alloys subject to ablative decomposition during atmospheric entry. This determination depends on the per-unit data identified in input (iv) below; without knowing what is reentering, the Commission cannot evaluate how much is reentering or what it deposits.

(iv) The per-unit satellite mass, physical dimensions, and material profile for each hardware variant the Applicant intends to deploy. The Technical Attachment states that "SpaceX plans to design and operate different versions of satellite hardware to optimize operations across orbital shells." TECH at A-1. The radiofrequency parameters filed in Schedule S are described as "representative." TECH at A-2. The Application does not state how many hardware variants will be deployed, how those variants differ in mass or material composition, or how their ablative behavior during atmospheric reentry may vary. Without this input, the Commission cannot evaluate either the per-event atmospheric signature or the aggregate atmospheric consequences of the described disposal architecture. The absence of per-unit physical dimensions also prevents evaluation of the optical and infrared brightness profile of the constellation at the described scale, given that satellite visibility is a function of physical size, solar array area, and orbital altitude. AAS Petition at 3 to 4.

(v) The cumulative atmospheric loading at the described disposal cadence, measured against the peer-reviewed baselines now on the record. See Murphy et al., 120 Proc. Nat'l Acad. Sci. e2313374120 (2023); Wing et al., 7 Comms. Earth & Env't 161 (2026); Maloney et al., 130 J. Geophys. Res. Atmos. e2024JD042442 (2025).

(vi) The launch cadence required for initial deployment and steady-state maintenance of the described constellation, and whether any existing environmental review of the Applicant's launch facilities encompasses that cadence.

The Applicant's own Waiver Requests acknowledge that key operational parameters "cannot be accurately described within the current structured format." WAIV. A filing vehicle that cannot encode the system's operative parameters cannot support the environmental certification that depends on those parameters. The Applicant is not a new entrant lacking operational experience from which to derive these inputs. The Technical Attachment states that SpaceX has been "operating over 10,000 satellites over the course of 7 years." TECH at A-11. An applicant with that operational history possesses the engineering data necessary to bound the lifecycle parameters of a successor system. The absence of that data from the record is not a function of unavailability. It is a function of non-disclosure. The Commission should not accept an unbounded environmental certification from an applicant that has the operational basis to provide a bounded one.

These are not peripheral details. They are the quantitative predicates on which any environmental determination must rest. A finding that the proposed action will not have a significant environmental effect is a factual conclusion. It requires factual inputs. Where the Application itself does not supply those inputs, and the record contains no independent basis on which the Commission could derive them, the environmental certification on Form 312 lacks the evidentiary foundation that Section 1.1307 presupposes.

The Petitioner does not ask the Commission to assume a particular answer to any of these questions. The Petitioner asks the Commission to require answers before making a determination that depends on them.

III. THE CENTRAL ENVIRONMENTAL PROBLEM: ATMOSPHERIC BURN-UP AS THE DISPOSAL MODEL

The Applicant offers three disposal pathways: atmospheric reentry, Earth disposal orbits, and heliocentric disposal orbits. NAR at 5. The Technical Attachment states: "Disposal orbit altitudes have not been selected." TECH at A-12. The record does not bound propellant budgets for non-atmospheric disposal and does not demonstrate that either alternative pathway is operationally feasible at the described scale. The record does not address whether the delta-v requirements for reaching heliocentric or Earth disposal orbits from operational altitudes of 500 to 2,000 km are propellant-feasible at the described scale, or whether those pathways would require propellant reserves that materially reduce the satellite's operational payload fraction. If they do, those pathways are not real alternatives. They are narrative alternatives, and the Commission's environmental analysis must proceed on the assumption that atmospheric burn-up is the predominant disposal method.

Atmospheric reentry appears to be the only disposal pathway the Applicant has committed to engineering. NAR at 5 ("will design its satellites to pose no calculable risk to humans upon re-entry and to minimize atmospheric impact"). If the design lifetime is approximately five years, as is typical of the Applicant's current Starlink hardware and as indicated on the filed Schedule S, a constellation of one million satellites yields a reasonable illustrative steady-state replacement rate of approximately 200,000 satellites per year. This figure is offered because the Application provides no bounded lifecycle-throughput envelope. The Amazon LEO Petition independently derives this figure and characterizes it as "more than 44 times the entire global satellite launch output in 2025, just to keep the constellation at steady state." Amazon LEO Petition at 10.

At 200,000 reentries per year, the system would produce approximately 547 reentry events per day, each involving the complete ablation of a spacecraft constructed from aerospace-grade materials and alloys the Application does not disclose. This is arithmetic applied to the Applicant's own description. Each reentry event represents an irreversible and irretrievable commitment of refined strategic materials. The aerospace-grade materials consumed in ablation cannot be recovered, recycled, or reused. The recovery rate is zero. The Application describes no recovery mechanism, no controlled-reentry collection capability, no on-orbit refurbishment or material-reclamation architecture, and no commitment to develop one. The Technical Attachment's reference to Earth disposal orbits and heliocentric disposal orbits does not constitute a recovery pathway. Parking a defunct satellite in a disposal orbit preserves the object but does not recover the material. Unless the Applicant demonstrates an operational architecture for retrieving, refurbishing, or recycling end-of-life spacecraft before they are committed to atmospheric ablation, every satellite disposed of through reentry represents a permanent, irrecoverable material loss. The Commission's environmental analysis should proceed on the basis of what the record contains, not on the basis of recovery capabilities the Applicant has neither described, committed to, nor demonstrated at any scale.

To the extent the Applicant contends that orbital storage preserves the option of future recovery, the burden is on the Applicant to identify the recovery vehicle, the operational architecture, the timeline, and the economic feasibility of retrieval at the described scale. An unfunded aspiration to collect debris at a future date is not a disposal pathway. It is a deferral of disposal with no demonstrated endpoint. The Commission cannot credit a recovery option that exists nowhere in the record as a basis for concluding that the irreversible commitment of strategic materials is mitigated. NEPA requires agencies to study, develop, and describe appropriate alternatives to proposed courses of action involving unresolved conflicts concerning alternative uses of available resources. 42 U.S.C. Section 4332(2)(E). The permanent atmospheric destruction of aerospace-grade strategic materials at industrial scale, when recovery or reuse alternatives have been neither evaluated nor foreclosed, is precisely such a conflict.

The full material composition of the proposed satellites is not in the record. The Applicant's own Narrative describes Starship as capable of deploying "millions of tons of mass per year to orbit when launching at rate." NAR at 4. At steady state, the deployment rate must equal the disposal rate. If atmospheric reentry is the predominant disposal method, as the record suggests, then the bulk of the mass launched must eventually reenter. Whatever the per-unit mass and material mix, the system's lifecycle architecture commits manufactured aerospace hardware to atmospheric destruction at industrial scale on a continuous, one-way basis.

NEPA identifies irreversible and irretrievable commitments of resources as a required dimension of environmental review. 42 U.S.C. Section 4332(2)(C)(v). NEPA separately requires discussion of alternatives to the proposed action, 42 U.S.C. Section 4332(2)(C)(iii), and the study of appropriate alternatives where a proposal involves unresolved conflicts concerning alternative uses of available resources, 42 U.S.C. Section 4332(2)(E). The irreversibility of a proposed action's resource commitments is also a recognized factor in determining whether the action may have a significant environmental effect warranting further review. The United States depends on foreign sources for a substantial share of the aerospace-grade alloys and strategic materials used in satellite manufacturing. A disposal architecture that permanently converts aerospace-grade materials to unrecoverable atmospheric particulate at industrial cadence would, within a few years of steady-state operation, represent a sustained strategic-material consumption stream of a magnitude the Commission has not previously authorized, with no recovery pathway, no substitution analysis, and no assessment of supply-chain implications anywhere in the record.

The present record contains no analysis of the cumulative material throughput, no assessment of whether the strategic materials consumed are subject to supply constraints or import dependencies, and no evaluation of whether alternative architectures could reduce the irretrievable material loss.

The Commission has never confronted the question of whether continual atmospheric deposition at this cadence, for the indefinite operational life of the system, constitutes a significant environmental effect within the meaning of Section 1.1307. The Applicant could resolve this question by disclosing the per-unit satellite mass, the expected disposal-method split, and the projected reentry cadence at each deployment phase. These are engineering parameters within the Applicant's possession. The Application's silence on these points is a choice, not a limitation.

IV. EVIDENCE OF POTENTIALLY SIGNIFICANT ENVIRONMENTAL EFFECTS ON THE RECORD

This Petition does not write on a blank slate. Multiple filings in the comment period have placed substantial evidence of potentially significant environmental effects before the Commission.

A. Atmospheric Contamination

Direct detection of spacecraft metals. Murphy et al. detected metals attributable to spacecraft reentry in stratospheric aerosol at concentrations above natural background, finding that approximately 10% of sulfuric acid particles larger than 120 nm already contain spacecraft-alloy metals. Murphy et al., 120 Proc. Nat'l Acad. Sci. e2313374120 (2023).

First time-resolved reentry measurement. Wing et al. measured a tenfold lithium enhancement at 96 km altitude from a single Falcon 9 upper-stage reentry, the first direct lidar observation linking a specific reentry event to a measurable atmospheric signature. Wing et al., 7 Comms. Earth & Env't 161 (2026). At the illustrative steady-state rate, this pathway would recur approximately 200,000 times annually.

Radiative forcing from reentry alumina. Maloney et al. modeled cumulative radiative forcing from reentry alumina at megaconstellation scales and found the forcing could become significant. Maloney et al., 130 J. Geophys. Res. Atmos. e2024JD042442 (2025).

Launch-emission climate effects. Maloney et al. found that stratospheric black carbon from rocket launches has a disproportionate radiative effect. The launch cadence required here would substantially exceed any modeled scenario. Maloney et al., 127 J. Geophys. Res. Atmos. e2021JD036373 (2022).

AAS atmospheric assessment. The AAS Petition states: "Without additional details, we cannot evaluate the effects of satellite disposal on the Earth's atmosphere, which have the potential to be significant, given the size and number of satellites involved." AAS Petition at 10. The AAS concluded that "a license should not be granted without a rigorous assessment of the potential environmental impacts." Id.

The Bakken IO independently placed the Murphy, Maloney, and Wing studies on the record and identified the absence of a bounded lifecycle-throughput envelope as a threshold deficiency. Bakken IO at 16, 24 to 25.

B. Night-Sky and Infrared Degradation

The AAS Petition presents simulations showing the constellation would produce tens of thousands of naked-eye-visible satellites globally, with "far more visible satellites than stars" for much of the globe, night, and year. AAS Petition at 3 to 5. Sun-synchronous orbits ensure sunlight exposure "up to more than 99% of the time." NAR at 3. The NSF-DOE Vera C. Rubin Observatory, representing over $800 million in federal investment, could become nearly inoperable. AAS Petition at 3 to 4. Artificial light at night could disrupt bird migration, pollination, and plant growth. AAS Petition at 9.

C. Radio-Frequency Interference

The AAS Petition demonstrates that aggregate emissions would exceed harmful interference thresholds in fourteen protected radio astronomy bands. Three interference pathways: Ka-band harmonics falling in protected bands from 76 through 174 GHz; aggregate spurious emissions scaling with constellation size; and unintended electromagnetic radiation from onboard electronics. Observational evidence of UEMR from existing Starlink satellites is confirmed. AAS Petition at 6 to 9.

D. Orbital-Environment Effects

The proposed constellation would increase the active satellite population by nearly 70 times over the total number of satellites now in orbit. DarkSky Petition at 4. The AAS further states that current models for existing large constellations already suggest a 40% chance of one or more ground casualties every five years, and that expanding the orbital population to one million units would likely elevate that casualty risk to a level the Commission has previously found unacceptable for smaller-scale deployments. AAS Petition at 11. The AAS derives the ground-casualty estimate from Wright, Boley, and Byers, who modeled collective casualty risks at megaconstellation scales. AAS Petition at 11, citing Wright, Boley & Byers, Space Policy 101749 (2026).

Amazon LEO likewise argues that managing collision risk at this scale would impose significant operational burdens on operators ascending, descending, or transiting through SpaceX's orbital shells, and that the proposed disposal approach could degrade the space environment while denying other operators access to new orbits at higher altitudes. Amazon LEO Petition at 7 to 8. These orbital-environment effects bear directly on Petitioner's direct-user injury identified in Section I.E.1: the proposed system's reliance on optical crosslinks routed through the existing Starlink constellation, at the scale described, creates a concrete risk that orbital congestion in the 500 to 2,000 km regime could degrade the network capacity on which Petitioner relies.

E. Launch Cadence

The Application proposes a constellation of up to one million satellites with a five-year design lifetime, implying a steady-state replacement rate of approximately 200,000 satellites per year. The Application does not disclose the per-unit satellite mass. Without that figure, the precise annual launch mass cannot be calculated, but any reasonable estimate yields a launch cadence far beyond existing environmental baselines. Amazon LEO's Petition to Deny independently derives the 200,000-satellite-per-year replacement rate and characterizes it as "more than 44 times the entire global satellite launch output in 2025, just to keep the constellation at steady state." Amazon LEO Petition at 10. In 2025, a record year, 4,526 satellites were launched worldwide. Id.

The Applicant's own Narrative describes Starship as capable of deploying "millions of tons of mass per year to orbit when launching at rate." NAR at 4. That description contextualizes the ODC program as dependent on a launch cadence that dwarfs any existing environmental review. The FAA's 2022 Programmatic Environmental Assessment for the SpaceX Starship/Super Heavy Launch Vehicle Program at the Boca Chica Launch Site analyzed up to five annual orbital Starship/Super Heavy launches and landings. FAA, Final Programmatic Environmental Assessment for the SpaceX Starship/Super Heavy Launch Vehicle Program at the SpaceX Boca

Chica Launch Site in Cameron County, Texas (June 2022); Mitigated FONSI/ROD issued June 13, 2022. The FAA's Tiered Environmental Assessment for SpaceX Starship/Super Heavy Vehicle Increased Cadence at the Boca Chica Launch Site analyzed up to 25 annual Starship/Super Heavy launches and 50 annual landings. FAA, Tiered Environmental Assessment for SpaceX Starship/Super Heavy Vehicle Increased Cadence at the SpaceX Boca Chica Launch Site in Cameron County, Texas. Even if each Starship launch carried hundreds of ODC satellites, deploying and replacing 200,000 satellites per year at steady state requires thousands of launches annually, orders of magnitude beyond the FAA's reviewed cadence.

The Boca Chica launch site is adjacent to the Lower Rio Grande Valley National Wildlife Refuge. The stratospheric black carbon effects of rocket launches at this cadence have not been modeled by any study in the peer-reviewed literature. Maloney et al. (2022) found disproportionate radiative effects from launch emissions at current global cadence; the implied ODC launch cadence would exceed the 2022 study's modeled scenarios by orders of magnitude.

The Commission's environmental determination for the ODC application cannot rest on the assumption that the FAA has reviewed or will review the launch-cadence environmental effects, because no FAA review at the implied cadence exists or has been initiated. If the Commission treats launch-cadence effects as outside its NEPA obligations, it should identify on the record which federal agency holds jurisdiction over the environmental effects of thousands of annual orbital launches from a site adjacent to a National Wildlife Refuge, and whether that agency has conducted review at the described scale.

V. WHY DARK-SKY DOES NOT FORECLOSE REVIEW

The D.C. Circuit's decision in Int'l Dark-Sky Ass'n v. FCC, 106 F.4th 1206 (D.C. Cir. 2024), evaluated the Commission's no-significant-impact determination for Gen2 Starlink: 7,500 satellites at lower altitudes. Three factors distinguish the present Application:

Scale. The proposed system is approximately 133 times larger. Environmental effects are not linear: atmospheric deposition scales with reentry count, optical brightness scales with visible objects, RF interference scales with aggregate emissions, and collision risk scales nonlinearly with density.

Altitude and visibility. Gen2 Starlink satellites at lower altitudes spend a larger fraction of their orbital period in Earth's shadow. The ODC application describes 500 to 2,000 km, including sun-synchronous orbits with 99%+ sunlight exposure. NAR at 3.

New science. The Dark-Sky record did not include Wing et al. (2026) or Maloney et al. (2025). These studies demonstrate that atmospheric effects of satellite reentry at scale are directly measurable and that cumulative radiative forcing at megaconstellation disposal cadences could be significant. The AAS Petition further places before the Commission simulation and interference data that was not part of the Dark-Sky record.

The Dark-Sky court did not hold that satellite constellations are immune from NEPA review. It held that the Commission's determination was reasonable on the record before it. A materially different record supports a different determination.

VI. THE ENVIRONMENTAL RECORD CANNOT SUPPORT THE APPLICATION'S CURRENT POSTURE

The Application proceeded without an Environmental Assessment notwithstanding the Section 1.1307 issues raised by the record. The Bakken IO argued that "the environmental certification on Form 312 states that a system requiring approximately 200,000 atmospheric reentries per year at steady state will have no significant environmental impact" and that "[p]ublished peer-reviewed research contradicts that assertion at the lifecycle throughput described." Bakken IO at 24.

The present record does not contain the six baseline inputs identified in Section II. The Application's environmental certification on Form 312 rests on those inputs being either established or unnecessary. They are neither. The Bureau should require the Applicant to supply them before any environmental determination, and should explain on the record the basis for any determination that proceeds without them. The environmental certification on Form 312 is made under 18 U.S.C. Section 1001. Where the record does not contain the inputs necessary to support a no-significant-impact conclusion, the question of how that certification was made and on what basis arises on the face of the filing. The Petitioner need not allege that the Applicant acted with intent to deceive. The Petitioner observes that the Applicant certified no significant environmental impact for a system whose environmental throughput the Application does not bound, using a filing vehicle the Applicant acknowledges cannot encode the system's operative parameters, notwithstanding seven years of operational data from a predecessor constellation that would permit a bounded showing. The Commission should require the Applicant to supply the bounded data it possesses before accepting a certification that depends on data the record does not contain.

To be clear: this Petition does not ask the Commission to develop new facts before deciding whether to act. The evidentiary basis for requiring further environmental review is already on the record. The peer-reviewed atmospheric science is on the record. Murphy et al. (2023), Wing et al. (2026), and Maloney et al. (2022, 2025). The interference analysis demonstrating exceedance of harmful thresholds in fourteen protected passive bands is on the record. AAS Petition at 6 to 9. The ground-casualty modeling is on the record. Wright, Boley & Byers (2026), as cited in AAS Petition at 11. The 70-fold increase in orbital population is on the record. DarkSky Petition at 4. The Applicant's own acknowledgment that its filing vehicle cannot accurately describe the proposed system is on the record. WAIV. The Applicant's own statement that disposal orbit altitudes have not been selected is on the record. TECH at A-12.

Section 1.1307(c) requires a written petition alleging that a categorically excluded action "may have a significant environmental effect." That standard is met when the record contains substantial evidence from which the Bureau could reasonably conclude that significant effects are possible. The record contains that evidence now. What the record does not contain are the Applicant's answers to the questions that evidence raises. Those answers are required for the Environmental Assessment the Bureau should direct the Applicant to prepare. They are not prerequisites to the Bureau's threshold determination that further environmental processing is warranted. The distinction matters: the six unresolved inputs identified in Section II are what the Applicant must supply in an EA, not what the Petitioner must supply to trigger one.

The categorical exclusion for satellite licensing was adopted in an era when constellations numbered in the dozens. It was not designed to absorb a system 133 times larger than any previously reviewed, implicating atmospheric, optical, radio-frequency, and orbital-environment effects documented by peer-reviewed science and the Commission's own record. Where the record before the Bureau contains the evidence identified above, the categorical exclusion cannot do the work the Applicant's Form 312 certification asks it to do. The Bureau's obligation under Section 1.1307(c) is to review the environmental concerns raised by this Petition. That obligation is ripe. The record is sufficient. The question is whether the Bureau will act on it.

VII. PRESERVED SUPPLEMENTAL THEORIES

The following theories are preserved in summary form. They are not necessary to support the core relief requested. The requested Environmental Assessment stands fully on the grounds in Sections II through VI.

A. Connected Actions and Anti-Segmentation

The ODC system cannot serve its described purpose independently of the separately licensed Starlink constellation. NAR at 6; TECH at A-1 to A-2. Under Delaware Riverkeeper Network v. FERC, 753 F.3d 1304, 1313 (D.C. Cir. 2014), connected actions must be considered together. The continuous replacement cycle is itself a connected action. The Petitioner notes that Seven County Infrastructure Coalition v. Eagle County, 605 U.S. 168 (2025), focuses on effects with a "reasonably close causal relationship" to the proposed action; the effects identified in this Petition flow directly from the system's design-for-demise architecture. Preserved.

B. Orbital Commons as Surface Features

Section 1.1307(a)(7) covers "significant change in surface features." One million satellites transforming narrow orbital shells from navigable commons to congestion-dominated environments is functionally analogous to the regulation's illustrative examples. Preserved.

C. Indian Religious Sites and the Night Sky

Section 1.1307(a)(5) covers facilities that "may affect Indian religious sites." The text does not limit "sites" to terrestrial locations. A constellation producing more visible satellites than stars would alter the medium through which Indigenous astronomical practices are conducted. Preserved.

D. RF Environment and Aggregate Exposure

Whether fleet-scale aggregate ground-level power flux density approaches the 5% threshold in Section 1.1307(b)(5)(i) is unresolved. The RF environment in protected passive bands may independently constitute a component of the "human environment" under NEPA. Preserved.

VIII. RELIEF REQUESTED

Petitioner respectfully requests that the Bureau and the Commission employ the appropriate procedural vehicle to ensure the environmental concerns raised herein are considered before any final action:

First, the Bureau should review this Petition and consider the environmental concerns raised, as directed by 47 C.F.R. Section 1.1307(c).

Second, the Bureau should determine that the Application may have a significant environmental impact.

Third, pursuant to 47 C.F.R. Sections 1.1307(c), 1.1308, and 1.1311, the Bureau should require the Applicant to submit an Environmental Assessment and defer action pending independent environmental review. The EA should be factual and concise, with sufficient detail to permit independent review, and should discuss unavoidable adverse effects, mitigation measures, and reasonable alternatives. The EA should also address whether alternatives to routine design-for-demise disposal were considered, including orbital recovery, refurbishment, servicing, modular replacement, longer-life architectures, or other approaches that could materially reduce the need for continual atmospheric ablation of spacecraft at the described scale; and whether the Commission's existing supervisory and enforcement mechanisms are adequate to ensure compliance with any environmental mitigation conditions at the scale described, or whether the absence of a commensurate supervisory framework itself constitutes a barrier to reasoned environmental review.

Petitioner does not assert that full spacecraft recovery or reuse is established on the present record; rather, Petitioner requests that the Applicant be required to address whether alternatives to routine atmospheric burn-up exist or could be incorporated, and whether current industry practice is being assumed rather than environmentally evaluated. If, after review, the Bureau determines the proposal will have a significant effect on the human environment, it should proceed under 47 C.F.R. Section 1.1314. To the extent the Bureau determines, on the basis of the evidence already in the record, that the significance threshold is met without further factual development, the Bureau should proceed directly under Section 1.1314 to require an Environmental Impact Statement without requiring an intermediate Environmental Assessment.

Fourth, to the extent further environmental processing is warranted, no grant should issue unless and until the Commission has completed the required environmental review.

Fifth, in the alternative, if the Bureau does not require an EA, it should at minimum issue a written determination addressing whether the action may have a significant environmental impact, identify any categorical exclusion relied upon, and explain why the record does not present extraordinary circumstances. This ensures a reviewable administrative record. See Motor Vehicle Mfrs. Ass'n v. State Farm Mut. Auto. Ins. Co., 463 U.S. 29, 43 (1983).

Sixth, before any environmental determination, the Bureau should require the Applicant to address the six unresolved inputs identified in Section II, the disposal-feasibility question developed in Attachment A, and whether alternatives to routine design-for-demise disposal could materially reduce reentry throughput and cumulative atmospheric loading at the scale described.

Seventh, the Petitioner requests leave to supplement this filing with additional scientific authorities, administrative notice materials, and applicant-response materials, and leave to correct any material error of fact or citation identified after filing, including by errata or amended filing, without waiver of the arguments or theories presented herein.

Eighth, to the extent the Bureau declines to process this filing under Section 1.1307(c), or declines to grant waiver, alternative treatment, or petition-to-deny status, the Bureau should state on the record the regulatory basis for that determination, including whether the ruling rests on timeliness, procedural vehicle, standing, lack of good cause, or some other identified ground. To the extent the Bureau disposes of the Application without separately addressing this Petition, the Petitioner requests that the order state whether the Bureau considered the environmental concerns raised herein and the basis on which it determined that further environmental processing was not required.

Ninth, to the extent this filing is treated concurrently as a petition to deny under 47 U.S.C. Section 309(d), the Commission should deny the Application on the grounds that grant would not serve the public interest, convenience, and necessity. The record before the Commission demonstrates that the Applicant has not provided the baseline technical and environmental data necessary for the Commission to make the findings required by Section 309(a). The Ownership Attachment does not reflect the current ownership and control picture following the February 2, 2026 merger. The filing vehicle the Applicant selected cannot, by the Applicant's own acknowledgment, accurately describe the proposed system. The environmental certification on Form 312 rests on inputs the record does not contain. The Application requests authority to deploy a system whose lifecycle-throughput architecture would produce atmospheric, optical, radio-frequency, and orbital-environment effects that multiple parties and peer-reviewed scientific authorities have identified as potentially significant, and the Applicant has not demonstrated that those effects are bounded, mitigated, or insignificant. Under these circumstances, the Application is not grantable on the present record. The Commission should deny the Application without prejudice to refiling with a complete record, or in the alternative, should designate the Application for hearing under 47 U.S.C. Section 309(e) on the environmental, ownership, and public-interest issues identified in this Petition and the Bakken IO.

Tenth, the Commission should address on the record whether the procedural barriers documented in Section I.E.4, including the separation of ICFS from the public-facing ECFS portal, the nonfunctional ecfs@fcc.gov email address, the misdirection by Commission accessibility staff, the undisclosed CORES credentialing requirement, the reported inoperability of the ICFS portal during the period immediately following the Public Notice, and the absence of publicly accessible guidance for pro se participation in satellite licensing proceedings, are consistent with the meaningful public participation that NEPA and the Commission's own rules contemplate. To the extent these barriers contributed to the absence of structured environmental and public-interest arguments from the early comment period, the Commission should state whether it considers the resulting record to reflect the informed public input the comment process is designed to produce, or whether supplemental comment opportunities or other corrective measures are warranted. This request is independently preserved regardless of the Commission's disposition of the environmental and public-interest claims raised herein.

IX. PRESERVATION

All environmental review theories identified in this Petition are preserved to the fullest extent available for any future administrative or judicial proceeding, including reconsideration, petition for review, and appeal. This includes the theories in Sections II through VI and the supplemental theories in Section VII.

This Petition is not waived by grant, partial grant, dismissal, designation for hearing, abeyance, or any other disposition. It is not waived by the Bureau's failure to separately adjudicate or reference this Petition in any order disposing of the Application. It is incorporated into any later petition for reconsideration or judicial review to the fullest extent permitted by 47 U.S.C. Sections 402(b) and 405, 28 U.S.C. Section 2342, and 5 U.S.C. Section 706. To the extent the Commission declines to consider this filing on any procedural ground, including timeliness, procedural vehicle, forum, format, standing, failure to state a claim, or any other basis that does not reach the substance of the environmental and public-interest concerns raised herein, Petitioner preserves the following: (1) that Section 1.1307(c) independently requires Bureau review of the environmental concerns raised by petition, without reference to the comment-period deadlines of Section 25.154, and that a refusal to review on timeliness grounds is inconsistent with the plain text of the regulation; (2) that the Supplemental Informal Objection under Section 25.154 supplements a timely-filed Informal Objection already on the record and must be considered before grant; (3) that the Commission's independent obligation under 47 U.S.C. Section 309(a) to evaluate the public interest before grant is not discharged by declining to consider a filing that places substantial evidence of environmental and public-interest concerns before the Bureau; (4) that a procedural refusal that does not address the substance of the environmental record is reviewable under the arbitrary and capricious standard of 5 U.S.C. Section 706(2)(A) and under Motor Vehicle Mfrs. Ass'n v. State Farm Mut. Auto. Ins. Co., 463 U.S. 29, 43 (1983); and (5) that to the extent the Commission's procedural rules do not accommodate a filing of this nature, the appropriate remedy is waiver or alternative treatment under 47 C.F.R. Section 1.3, not dismissal of substantial environmental concerns from the administrative record. A procedural disposition that leaves the environmental record unaddressed does not extinguish the Commission's NEPA obligations. It defers them, and the deferral is preserved for review.

Nothing in this Petition concedes that the Application is otherwise acceptable for filing or grantable on non-environmental grounds. The Bakken IO raised threshold defects, record omissions, and public-interest issues independent of and additional to the environmental concerns raised herein.

A. The Evolving NEPA Framework

The federal environmental review framework is in transition. The Council on Environmental Quality rescinded its government-wide NEPA implementing regulations effective April 11, 2025. 90 Fed. Reg. 10610 (Feb. 25, 2025). The FCC's own NEPA regulations at 47 C.F.R. Part 1, Subpart I, cross-reference CEQ's regulations at Section 1.1302. The rescission of the cross-referenced framework does not extinguish the FCC's independent NEPA obligations. NEPA itself, 42 U.S.C. Section 4332, has not been repealed. The Commission's Part 1 environmental rules remain in effect and are the operative framework for this proceeding unless and until the Commission amends them through notice-and-comment rulemaking.

The Commission's NEPA modernization rulemaking provides an appropriate vehicle for establishing environmental review procedures commensurate with megaconstellation-scale applications. To the extent the Commission concludes that its existing Part 1 environmental rules are inadequate to evaluate a system of this scale, the appropriate response is rulemaking, not case-by-case accommodation that effectively exempts the largest proposed system from the most rigorous review.

The Petitioner preserves: (1) that the categorical exclusion should be reconsidered for megaconstellation-scale applications; (2) that NEPA, 42 U.S.C. Section 4332, has not been repealed and the statutory obligation persists; (3) the proximate-causation and jurisdictional arguments set forth in the following paragraph; (4) that the Fiscal Responsibility Act amendments to NEPA, including the mandatory deadline and page-limit provisions at 42 U.S.C. Section 4336a, apply to any Environmental Assessment required in this proceeding and do not excuse the obligation to prepare one where the record supports it; (5) that the legality of the

Bureau's action should be judged under the governing statute and rules in effect when the agency acts; and (6) that Loper Bright Enterprises v. Raimondo, 144 S. Ct. 2244 (2024), requires a reviewing court to exercise independent judgment on questions of law, including whether the Commission's application of its NEPA regulations to this proceeding was legally correct, without deference to the agency's interpretation of those regulations.

The proximate-causation standard articulated in Seven County Infrastructure Coalition v. Eagle County, 605 U.S. 168 (2025), does not insulate this Application. The environmental effects identified in this Petition bear a reasonably close causal relationship to the Commission's licensing action because: (a) atmospheric reentry at end of life is a structural feature of the licensed system's stated disposal architecture; (b) optical and infrared visibility consequences arise directly from the orbital shells, inclinations, and sunlight-exposure profiles the Application requests authority to occupy; (c) radio-frequency effects arise from the communications authority sought in this Application and from aggregate emissions at the described fleet scale; and (d) orbital-environment transformation arises from the density and distribution of spacecraft the Application proposes to place in the 500 to 2,000 km regime. To the extent the Commission determines that any category of environmental effect identified in this Petition falls outside the scope of its NEPA obligations, the Petitioner requests that the Commission identify on the record which federal agency holds jurisdiction over that effect and whether that agency has conducted environmental review at the scale described in the Application.

If the Commission processes this Application without requiring environmental review commensurate with its described scale, that disposition will establish the practical baseline for every future megaconstellation application. The Commission should consider whether the environmental review posture it adopts here is one it is prepared to apply to any applicant proposing a system of comparable or greater magnitude.

No argument is waived by its summary treatment herein or by its non-development in the Bakken IO.

Respectfully submitted,

Nickolai G. Bakken

/s/ Nickolai G. Bakken

Pro Se

Risaralda, Colombia

VERIFICATION / DECLARATION

I, Nickolai G. Bakken, declare under penalty of perjury under the laws of the United States of America that I am the Petitioner in the foregoing Petition for Environmental Review and Attachment A. I am a citizen of the United States whose official address of record is in Treasure Island, Florida, and I am currently traveling internationally in Risaralda, Colombia. I have read the foregoing Petition and Attachment A in their entirety. The factual statements contained therein concerning my personal identity, citizenship, residence, Starlink terminal ownership, intermittent service usage, night-sky observational practices, travel to observe celestial events, and AI-assisted creative and educational projects involving astrophysical data are true and correct to the best of my personal knowledge. Scientific literature, government reports, docket filings, and other record materials cited therein are presented for the consideration of the Commission and any reviewing body, and for record-preservation purposes, and are accurately characterized to the best of my knowledge and belief. Legal arguments, technical analysis, and inferences drawn from the administrative record are offered as argument, not as matters of personal knowledge.

Executed on 03/09/2026.

/s/ Nickolai G. Bakken
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CERTIFICATE OF SERVICE

I, Nickolai G. Bakken, hereby certify that on 03/09/2026, I caused a copy of the foregoing Petition for Environmental Review and Supplemental Informal Objection, together with its Attachment A, to be served by electronic mail upon the following:

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